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Introduction

Project Overview

The project centers on a comprehensive analysis of Olist, a Brazilian e-commerce platform. Olist operates as a marketplace connecting small to medium-sized businesses to customers across Brazil. The project utilizes a rich, real-world dataset encompassing over 100,000 orders, 3,000+ sellers, and 30,000+ products from 2016 to 2018.

Business Context

Olist is a Brazilian e-commerce marketplace that connects small and medium-sized businesses with customers across the country. It operates within a highly competitive and logistically complex environment, characterized by vast geographic distances and diverse regional economics. This analysis focuses on understanding the operational dynamics, customer base, and sales performance of this marketplace to drive strategic decision-making and sustainable growth.

Project Objective

- Tracking order statuses from purchase to delivery.
- Profiling customer geography, purchasing behavior, and retention rates.
- Analyzing sales revenue, category performance, and inventory activity.
- Mapping customer and seller concentration across Brazil.
- Identify sales growth patterns, seasonal fluctuations, and key business milestones.

Tools & Technology

- Database Management: MySQL for data warehousing and query execution.
- Data Visualization: PowerBI for interactive dashboards and executive reporting.
- Data Modeling: Star schema with fact and dimension tables.

Methodology

Analytical Framework

The project employs a structure multi-phase combining data engineering ,statistical analysis and business intillegence techniques to derive actionable insights from automative sales data.

Data Collection and Profiing

- Dataset Collected from kaggle dataset
- Data Profiling
 - Uniqueness Analysis : Validation of primary integrity across all tables
 - Cardinality Assessment : Evaluation of distinct value counts and uniqueness percentage
 - Data type Classification : Categorization of fields into numerical ,categorical ,temporal and text types
 - Completenes Chack: Identification of null values and missing data patterns

Analytical Techniques

- Summery Statistics
- Aggregation & Grouping
- Trend Analysis
- Pareto Analysis(80/20 Rule)
- Cohort Analysis
- Market Concentration Measuremnt
- Key Performance Indicator
- ABC Product Classification
- Ranking
- Segementation
- Transition Matrix
- Cumulative Pattern

Statistical Method

- Estimates of Location
 - Mean
 - median
 - Outlier
- Estimates of Variability :
 - Variance,
 - Standard deviation
 - Range ,Percentile

Data Schema & Modification

Pre-Normalization Data Schema

- Customers

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
customer_id	text	99441	99441	Unique Identifier	100.00%
customer_unique_id	text	96096	99441	Unique Identifier	96.64%
customer_zip_code_prefix	bigint	14994	99441	Numerical Continuous	15.08%
customer_city	text	4119	99441	Nominal / Text (High Cardinality)	4.14%
customer_state	text	27	99441	Nominal / Text (High Cardinality)	0.03%

- Geolocation

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
geolocation_lat	double	717360	1000163	Numerical Continuous	71.72%
geolocation_lng	double	717613	1000163	Numerical Continuous	71.75%
geolocation_zip_code_prefix	bigint	19015	1000163	Numerical Continuous	1.90%
geolocation_city	text	5969	1000163	Nominal / Text (High Cardinality)	0.60%

geolocation_state	text	27	1000163	Nominal / Text (High Cardinality)	0.00%
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▪ Order_items

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
freight_value	double	6999	112650	Numerical Continuous	6.21%
order_item_id	bigint	21	112650	Numerical Continuous	0.02%
price	double	5968	112650	Numerical Continuous	5.30%
order_id	text	98666	112650	Nominal / Text (High Cardinality)	87.59%
product_id	text	32951	112650	Nominal / Text (High Cardinality)	29.25%
seller_id	text	3095	112650	Nominal / Text (High Cardinality)	2.75%
shipping_limit_date	text	93318	112650	Nominal / Text (High Cardinality)	82.84%

▪ Order_payments

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
order_id	text	99440	103886	Unique Identifier	95.72%
payment_installments	bigint	24	103886	Numerical Continuous	0.02%
payment_sequential	bigint	29	103886	Numerical Continuous	0.03%
payment_value	double	29077	103886	Numerical Continuous	27.99%
payment_type	text	5	103886	Nominal / Categorical	0.00%

▪ Order_reviews

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
order_id	text	98673	99224	Unique Identifier	99.44%
review_answer_timestamp	text	98248	99224	Unique Identifier	99.02%
review_id	text	98410	99224	Unique Identifier	99.18%
review_score	bigint	5	99224	Numerical Discrete	0.01%
review_comment_message	text	35431	99224	Nominal / Text (High Cardinality)	35.71%
review_comment_title	text	3901	99224	Nominal / Text (High Cardinality)	3.93%
review_creation_date	text	636	99224	Nominal / Text (High Cardinality)	0.64%

▪ Orders

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
customer_id	text	99441	99441	Unique Identifier	100.00%
order_delivered_customer_date	text	95664	99441	Unique Identifier	96.20%
order_id	text	99441	99441	Unique Identifier	100.00%
order_purchase_timestamp	text	98875	99441	Unique Identifier	99.43%
order_status	text	8	99441	Nominal / Categorical	0.01%
order_approved_at	text	90733	99441	Nominal / Text (High Cardinality)	91.24%
order_delivered_carrier_date	text	81018	99441	Nominal / Text (High Cardinality)	81.47%
order_estimated_delivery_date	text	459	99441	Nominal / Text (High Cardinality)	0.46%

- Product_category_name_translation

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
product_category_name	text	71	71	Unique Identifier	100.00%
product_category_name_english	text	71	71	Unique Identifier	100.00%

- Products

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
product_id	text	32951	32951	Unique Identifier	100.00%
product_description_lenght	double	2960	32951	Numerical Continuous	8.98%
product_height_cm	double	102	32951	Numerical Continuous	0.31%
product_length_cm	double	99	32951	Numerical Continuous	0.30%
product_name_lenght	double	66	32951	Numerical Continuous	0.20%
product_weight_g	double	2204	32951	Numerical Continuous	6.69%
product_width_cm	double	95	32951	Numerical Continuous	0.29%
product_photos_qty	double	19	32951	Numerical Discrete	0.06%
product_category_name	text	73	32951	Nominal / Text (High Cardinality)	0.22%

- Sellers

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
seller_id	text	3095	3095	Unique Identifier	100.00%
seller_zip_code_prefix	bigint	2246	3095	Numerical Continuous	72.57%
seller_city	text	610	3095	Nominal / Text (High Cardinality)	19.71%

seller_state	text	23	3095	Nominal / Text (High Cardinality)	0.74%
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Table Size in Schema

Table	Size (MB)
geolocation	66.59
orders	21.55
order_items	20.55
order_reviews	17.55
customers	12.52
order_payments	10.52
products	4.52
sellers	0.28
product_category_name_translation	0.02

Normalization & Data Modeling

Data Model Schema

▪ dim_customer

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
customer_id	varchar	99163	99163	Unique Identifier	100.00%
customer_unique_id	varchar	95828	99163	Unique Identifier	96.64%
customer_city	varchar	4074	99163	Nominal / Text (High Cardinality)	4.11%
customer_state	varchar	27	99163	Nominal / Text (High Cardinality)	0.03%
customer_zip_code_prefix	varchar	14837	99163	Nominal / Text (High Cardinality)	14.96%

▪ dim_customer_geo

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
zip_code_prefix	varchar	19015	19015	Unique Identifier	100.00%
geolocation_lat	decimal	18987	19015	Numerical Continuous	99.85%
geolocation_lng	decimal	18988	19015	Numerical Continuous	99.86%
geolocation_city	varchar	5771	19015	Nominal / Text (High Cardinality)	30.35%
geolocation_state	varchar	27	19015	Nominal / Text (High Cardinality)	0.14%

▪ dim_date

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
date_key	int	634	634	Numerical Continuous	100.00%
day	int	31	634	Numerical Continuous	4.89%
week_of_year	int	52	634	Numerical Continuous	8.20%
is_weekend	tinyint	2	634	Numerical Discrete	0.32%
month	int	12	634	Numerical Discrete	1.89%
quarter	int	4	634	Numerical Discrete	0.63%
year	int	3	634	Numerical Discrete	0.47%
full_date	date	634	634	Temporal	100.00%
day_name	varchar	7	634	Nominal / Categorical	1.10%
month_name	varchar	12	634	Nominal / Categorical	1.89%

▪ dim_order_payments

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
order_id	varchar	99440	103886	Unique Identifier	95.72%

payment_id	int	103886	103886	Numerical Continuous	100.00%
payment_installments	int	24	103886	Numerical Continuous	0.02%
payment_sequential	int	29	103886	Numerical Continuous	0.03%
payment_value	decimal	29077	103886	Numerical Continuous	27.99%
payment_type	varchar	5	103886	Nominal / Categorical	0.00%

▪ dim_order_review

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
order_id	varchar	98673	99224	Unique Identifier	99.44%
review_id	varchar	98410	99224	Unique Identifier	99.18%
review_score	int	5	99224	Numerical Discrete	0.01%
review_answer_timestamp	datetime	98248	99224	Temporal	99.02%
review_creation_date	datetime	636	99224	Temporal	0.64%
review_comment_message	text	35431	99224	Nominal / Text (High Cardinality)	35.71%
review_comment_title	varchar	3901	99224	Nominal / Text (High Cardinality)	3.93%

▪ dim_product

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
product_id	varchar	2045	2045	Unique Identifier	100.00%
product_description_length	int	1094	2045	Numerical Continuous	53.50%
product_height_cm	decimal	77	2045	Numerical Continuous	3.77%
product_length_cm	decimal	88	2045	Numerical Continuous	4.30%
product_name_length	int	53	2045	Numerical Continuous	2.59%

product_weight_g	decimal	474	2045	Numerical Continuous	23.18%
product_width_cm	decimal	62	2045	Numerical Continuous	3.03%
product_photos_qty	int	16	2045	Numerical Discrete	0.78%
price_class	varchar	0	2045	Nominal / Categorical	0.00%
product_category_name	varchar	7	2045	Nominal / Categorical	0.34%

▪ dim_product_category_name_translation

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
product_category_name	varchar	71	71	Unique Identifier	100.00%
product_category_name_english	varchar	71	71	Unique Identifier	100.00%

▪ dim_seller

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
seller_id	varchar	3088	3088	Unique Identifier	100.00%
seller_city	varchar	610	3088	Nominal / Text (High Cardinality)	19.75%
seller_state	varchar	23	3088	Nominal / Text (High Cardinality)	0.74%
seller_zip_code_prefix	varchar	2239	3088	Nominal / Text (High Cardinality)	72.51%

▪ dim_seller_geo

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
zip_code_prefix	varchar	19015	19015	Unique Identifier	100.00%
geolocation_lat	decimal	18987	19015	Numerical Continuous	99.85%
geolocation_lng	decimal	18988	19015	Numerical Continuous	99.86%

geolocation_city	varchar	5771	19015	Nominal / Text (High Cardinality)	30.35%
geolocation_state	varchar	27	19015	Nominal / Text (High Cardinality)	0.14%

▪ fact_order_items

Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
freight_value	decimal	6999	112650	Numerical Continuous	6.21%
item_id	int	21	112650	Numerical Continuous	0.02%
price	decimal	5968	112650	Numerical Continuous	5.30%
shipping_limit_date	datetime	93318	112650	Temporal	82.84%
order_id	varchar	98666	112650	Nominal / Text (High Cardinality)	87.59%
product_id	varchar	32951	112650	Nominal / Text (High Cardinality)	29.25%
seller_id	varchar	3095	112650	Nominal / Text (High Cardinality)	2.75%

▪ fact_orders

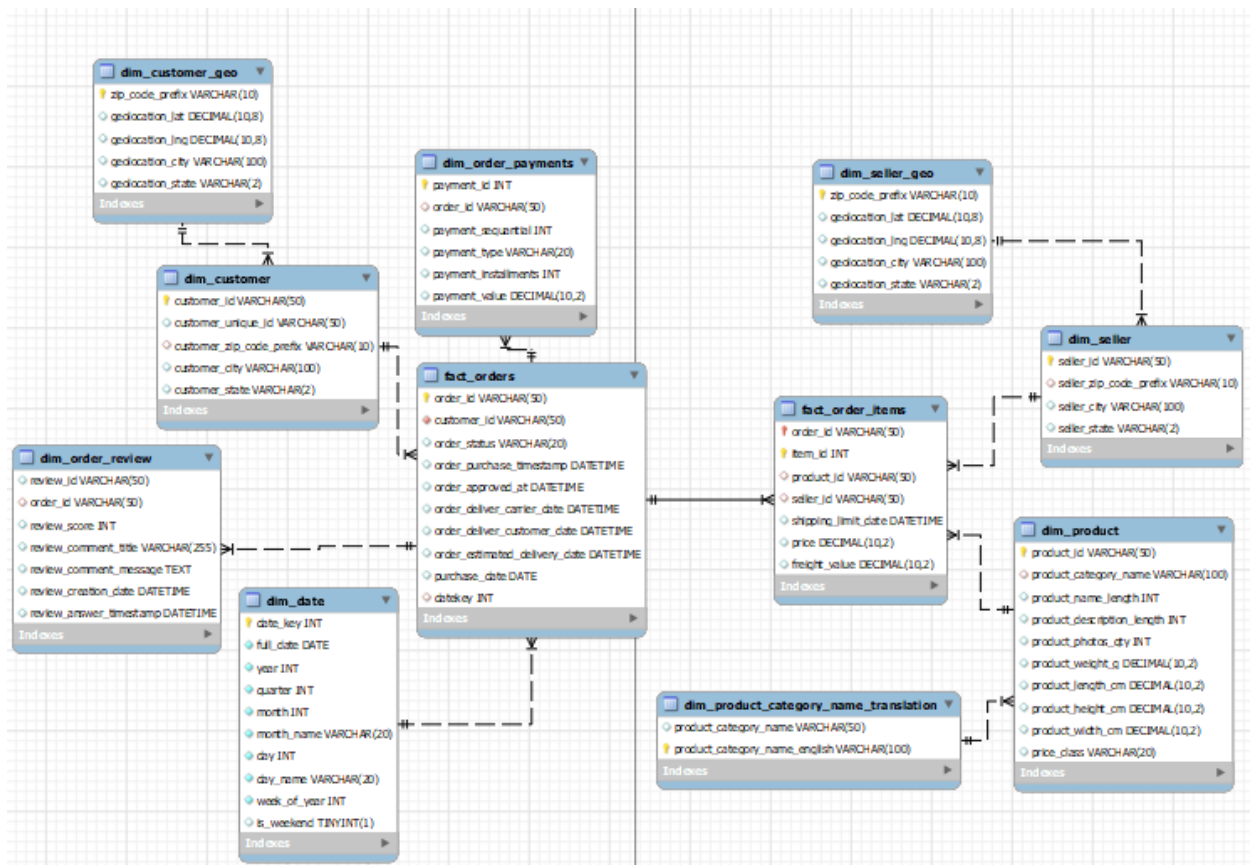
Column Name	Data Type	Distinct Values	Total Rows	Category	Uniqueness %
customer_id	varchar	99163	99163	Unique Identifier	100.00%
order_id	varchar	99163	99163	Unique Identifier	100.00%
datekey	int	634	99163	Numerical Continuous	0.64%
order_approved_at	datetime	90498	99163	Temporal	91.26%
order_deliver_carrier_date	datetime	80826	99163	Temporal	81.51%
order_deliver_customer_date	datetime	95404	99163	Temporal	96.21%
order_estimated_delivery_date	datetime	459	99163	Temporal	0.46%
order_purchase_timestamp	datetime	98601	99163	Temporal	99.43%
purchase_date	date	0	99163	Temporal	0.00%

order_status	varchar	8	99163	Nominal / Categorical	0.01%
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■ Table size schema

Table	Size (MB)
fact_order_items	47.84
fact_orders	32.69
dim_customer	20.11
dim_customer_geo	2.52
dim_seller_geo	2.52
dim_seller	0.52
dim_product	0.39
dim_date	0.06
dim_order_payments	0.03
dim_order_review	0.03
dim_product_category_name_translation	0.02

ERD Diagram



Feature Engineering Insight

- Order Approved time
- Delivery Time span
- Carrier handoff time
- Item value
- Total order value
- Freight percentage
- Customer behavior
- Installment category
- Customer location
- Seller performance
- Customer satisfaction
- Customer distance to seller

Identifier Based Analysis

Order Identifier Section

Order Status

order_status	order_count	percentage	running_pct
delivered	96214	97.03	97.03
shipped	1100	1.11	1.11
canceled	623	0.63	0.63
unavailable	607	0.61	0.61
invoiced	313	0.32	0.32
processing	300	0.30	0.30
created	4	0.00	0.00
approved	2	0.00	0.00

Key Insight

- 97.29% fulfilment industry benchmark
- Very good cancellation rate below then 2-5%
- Operationla flowalso healthy
- Unabilble 0.61%

Business Actions

- Investigate root cause of 609 unabailake orders
- Monitortime spent in "Invoice" and "processing" stages

Order Level Percentage

year	quarter	month	month_name	order_count	quarter_avg	pct_of_year
2016	3	9	September	4	4	1.22
2016	4	10	October	324	163	98.48
2016	4	12	December	1	163	0.30
2017	1	1	January	800	1751	1.78
2017	1	2	February	1779	1751	3.96
2017	1	3	March	2674	1751	5.95
2017	2	4	April	2398	3106	5.33
2017	2	5	May	3689	3106	8.20
2017	2	6	June	3232	3106	7.19
2017	3	7	July	4017	4203	8.93
2017	3	8	August	4319	4203	9.60

2017	3	9	September	4273	4203	9.50
2017	4	10	October	4617	5932	10.27
2017	4	11	November	7521	5932	16.72
2017	4	12	December	5657	5932	12.58
2018	1	1	January	7255	7049	13.47
2018	1	2	February	6704	7049	12.45
2018	1	3	March	7189	7049	13.35
2018	2	4	April	6918	6641	12.84
2018	2	5	May	6855	6641	12.73
2018	2	6	June	6151	6641	11.42
2018	3	7	July	6275	4261	11.65
2018	3	8	August	6491	4261	12.05
2018	3	9	September	16	4261	0.03
2018	4	10	October	4	4	0.01

Business Insight

- This business showed phenomenal growth in 2017 but faced major disruption in late 2018 that needs urgent investigation .

Yearly Growth

year	Q1 order	Q2 order	Q3 order	Q4 order	Total	YoY Growth %
2016	0	0	4	325	329	No Data
2017	5253	9319	12609	17795	44976	13570.52
2018	21148	19924	12782	4	53858	19.75

Business Insight

- 2017 experienced explosive growth (13,570% YOY), indicating rapid marketplace expansion and strong customer/seller onboarding
- 2018 maintained positive growth(19.75% YOY),showing the business began to stabilize after the hyper-growth phase.
- Q1 and Q2 of 2018 were strong,but Q3 lowed ,and Q4 orders collapsed suggesting a major disrupting data.
- Growth transition from 2017 to 2018 reflects a shift from hyper-scaling to market maturity.
- Q4 2018 anomaly requires immediate investigation into operational or data pipeline issues.

Monthly Growth

Year	January	February	March	April	May	June	July	August	September	October	November	December	total	YOY Growth %
2016	0	0	0	0	0	0	0	0	4	324	0	1	329	
2017	800	1779	2674	2398	3689	3232	4017	4319	4273	4617	7521	5657	44976	13570.52
2018	725	6704	7189	6918	6855	6151	6275	6491	16	4	0	0	5385	19.75

Business Insight

- 2017 shows strong ,consistent month-over-month growth,confirming rapid customer acquisition and marketplace scaling throughout the year.
- Peak demand months in 2017 were November and December in 2017 were November and December ,driven by seasonal sales(Black Friday,holiday period).
- 2018 starts with very high monthly volumes ,especially in January-March indicating strong Indicating strong returning customer momentum and sustained demand
- From April to August 2018 ,sales remain stable but show slight month-to-month decline,suggesting early signs of market saturation or operational limits.
- September-December 2018 collapse to near-zero orders ,signaling a critical business or data disruption ,not a natural trend
- Overall,2018 still shows +19.75% YOY growth,but the Q4 anomaly requires immediate investigation to avoid misinterpretation of business performance.

Monthly in Day wise

month_name	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Total
January	73	204	225	255	242	220	8055
February	304	278	253	267	346	350	8483
March	375	336	287	316	337	374	9863
April	275	347	318	353	361	284	9316
May	373	423	406	372	307	320	10544
June	308	266	280	326	347	359	9383
July	241	303	333	381	339	275	10292
August	473	457	462	387	391	479	10810

September	157	102	129	151	158	142	4293
October	128	144	207	219	187	179	4945
November	111	124	142	111	142	192	7521
December	274	216	233	335	282	278	5658

Business Insight

- August ,May and July are the highest performing months,indicating strong seasonal demand and possibly major promotional campaigns.
- September and October show sharp drops,signaling slowdown or external disruptions affecting order volume.
- Across all months,Thursday and Friday consistently show strong activity ,suggesting these are peak purchase days for customers.
- Weekends(Saturday) maintain stable order volumes but are generally lower than weekdays ,reflecting typical e-commerce behavior where buyers prefer weekdays for online purchases.
- Monday is the weakest day in most months,indicating slower post-weekend engagement
- The high weekday consistency (Tuesday-Friday) suggests predictable customer shopping patterns ,useful for logistics and marketing planning.
- The spike in August across almost all weekdays points toward a successful seasonal or promotional cycle during that period.

Return Customer

order_sequence	order_count	customer_count	customer_percentage
First_Order	95828	95828	100.00
Second_Order	2988	2988	3.12
Third_Order	251	251	0.26
4-5_th_Order	68	49	0.05
6+_Orders	28	11	0.01

Business Insight

- The business has an extremely high proportion of one time customers(100% are first-order customers),indicating low retention and limited repeat engagement.
- Only 3.12% of customers return for a second purchase,showing weak customer loyalty and limited long-term relationship building.
- Repeat purchase rates drop sharply after the second order,with third -order customers at just 0.26% ,and 4+ orders becoming extremely rare(<0.1%)
- This pattern decline from 1st – 2nd -3rd orders suggests the platform relies heavily on new customer acquisition ,not on retaining existing users.

- Improving customer experience ,delivery reliability ,and loyalty programs could significantly increase repeat purchases and overall lifetime value(LTV)

Days between orders

repurchase_window	customer_count	avg_days	pct_of_repaet_customers
0-30 days	1455	5.4	48.69
31-90 days	534	56.6	17.87
91-180 days	437	132.5	14.63
+180 days	562	292.8	18.81

Business Insight :

- Nearly 50% of repeat customers return within days ,indicating strong early satisfaction and product relevance among a small loyal segment
- Short repurchase window(0-30 days) with an average of 5.4 days shows customers who do return tend to purchase again quickly.
- 31-90 day and 91-180 day windows together account for about 32% of repeat customers return after 180+ days ,showing long-tail engagement but weak ongoing retention strategis.
- A significant 18.81% of repeat customers return after 180+days ,howing long-tail engagement but weak ongoing retention strategis.
- The increasing average days across windows suggest the business to maintain consistent engagement over time
- Overall ,the data highlights a high churn rate ,and improving customer retention programs could shift more users into the early repurchase window.

Customer Identifier Section

General Information Customer

total_cust omers	unique_indi viduals	duplicate_custo mer_ids	duplicate_ra te_pct	states_co vered	cities_co vered	zip_codes_ overed
99163	95828	3335	3.36	27	4074	14837

Business Insight

- The platform serves 99,163 customers, of which 95,828 are unique individuals, showing strong national-level penetration.
- Only 3.36% of customers' IDs are duplicates, indicating excellent data quality and reliable customer identity management.
- With customers spread across 27 states and 4,074 cities, the marketplace demonstrates broad geographic coverage and strong nationwide reach.
- The coverage of 14,837 unique ZIP code areas shows deep penetration even in smaller, long-tail locations.
- This widespread distribution highlights a scalable logistics network and the platform's ability to attract customers far beyond major metropolitan regions.
-

Investigate duplicate customer

ids_per_customer	customer_count	avg_zip_code	pct_of_duplicates
2	2737	1.08	91.60
3	202	1.11	6.76
4	30	1.07	1.00
5	8	1.00	0.27
6	6	1.17	0.20
7	3	1.00	0.10
9	1	3.00	0.03
17	1	1.00	0.03

Business Insight

- Most duplicates come from customers with exactly 2 IDs (91.6%), indicating minor inconsistencies rather than widespread data.
- Higher-order duplicates (3-7 IDs) are extremely rare, suggesting the duplication issue is localized and not systemic.

- The avg_zip_code ~1.0 across all groups shows that even duplicates IDs are tied to the same location, reinforcing that the issue is likely
 - ✓ account duplication
 - ✓ multiple registrations
 - ✓ system-generated ID inconsistencies
- Very high ID counts(9 and 17 IDs) are outliers ,likely caused by
 - ✓ Migration issues
 - ✓ Legacy system merges
 - ✓ Manual data entry erros
- Overall,duplicates represents only a small fraction of the customer base,meaning data quality is strong and deduplication is manageable.

Customer distribution by pareto analysis

customer _state	customer _count	pct_of_cus tomers	cumulative_ customers	cumulative_ pct	abc_category
SP	40287	42.02	40287	42.02	Pareto A(top 80%)
RJ	12372	12.91	52659	54.93	Pareto A(top 80%)
MG	11248	11.73	63907	66.66	Pareto A(top 80%)
RS	5273	5.50	69180	72.16	Pareto A(top 80%)
PR	4871	5.08	74051	77.24	Pareto A(top 80%)
SC	3534	3.69	77585	80.93	Pareto B(Next 15%)
BA	3268	3.41	80853	84.34	Pareto B(Next 15%)
ES	1959	2.04	82812	86.38	Pareto B(Next 15%)
GO	1944	2.03	84756	88.41	Pareto B(Next 15%)
DF	1908	1.99	86664	90.40	Pareto B(Next 15%)
PE	1605	1.67	88269	92.07	Pareto B(Next 15%)
CE	1310	1.37	89579	93.44	Pareto B(Next 15%)
PA	946	0.99	90525	94.43	Pareto B(Next 15%)
MT	875	0.91	91400	95.34	Pareto C(last 5%)
MA	722	0.75	92122	96.09	Pareto C(last 5%)
MS	694	0.72	92816	96.82	Pareto C(last 5%)
PB	517	0.54	93333	97.36	Pareto C(last 5%)
PI	479	0.50	93812	97.86	Pareto C(last 5%)
RN	472	0.49	94284	98.35	Pareto C(last 5%)
AL	400	0.42	94684	98.77	Pareto C(last 5%)
SE	341	0.36	95025	99.12	Pareto C(last 5%)

TO	272	0.28	95297	99.41	Pareto C(last 5%)
RO	238	0.25	95535	99.65	Pareto C(last 5%)
AM	143	0.15	95678	99.80	Pareto C(last 5%)
AC	77	0.08	95755	99.88	Pareto C(last 5%)
AP	67	0.07	95822	99.95	Pareto C(last 5%)
RR	45	0.05	95867	100.00	Pareto C(last 5%)

Business Insight

- Customer demand is heavily concentrated in a few key states – six states (SP,RJ,MG,RS,PR,PC) make up over 80% of the entire customer base ,clearly defining the core market region.
- Sao Paulo (SP) alone contribute 42% of all customers ,making it the dominant state and the primary revenue engine for the business
- RJ and MG together push cumulative share above 66% indicating strong customer cluster across Brazil's southeast region
- States in Pareto B(next 15%),such as BA,ES,GO,and DF represent secondary but high potential growth regions.Focused marketing could significantly increase penetration here.
- Pareto C states(last 5%),mostly northern and smaller states ,show long-tail,low-volume demand ,meaning expansion requires careful cost-benefit analysis due to logistic complexity.
- The strong skew toward a few state reveals a highly uneven market distribution ,suggesting strategic opportunities in growing under-penetrated regions while strengthening dominance in core areas.

HHI INDEX for market concentration

hhi_score

2173

concentration_risk

Moderate Concentration

Business Insight

- The HHI score of 2173 indicates a moderately concentrated customer market, meaning a few states dominate the majority of demand ,'
- This suggests that while the platform has nationwide reach,customer activity is not evenly distributed across Brazil .
- The Business is dependent on handful of high-performing regions,especially ,SP,RJ,and MG ,which increase regional risk exposure.
- At the same time ,a moderate HHI means there is significant room for expansion into under-penetrated states without facing monopolistic barriers.
- Strategic marketing and logistic investment in Tier-2 states could reduce concentration risk and unlock new growth opportunities.

City segmentation with long tail

city_tier	city_count	total_customers	avg_customers_per_city	pct_of_customers
top 10 cities	10	33631	3363.1	35.05
Top 11-50 cities	40	16961	424.0	17.68
Top 51-200 cities	150	18687	124.6	19.48
Long Tail(201+)	4059	26671	6.6	27.80

Business Insights

- The top 10 cities contribute 35% of total customers, showing extremely strong clustering in major metropolitan areas
- Cities ranked 11-50 contribute another 17.7%, meaning the top 50 cities alone account for over half (52%) of customer demand
- The long tail of 4,059 smaller cities contributes 27.8% confirming that the platform successfully reaches remote and low-density markets despite lower per-city volumes.
 - ✓ The average customer per city shows clear hierarchy
 - ✓ Top 10 cities - 3363 customers per city
 - ✓ Top 51-200 - only 125 per city

Top 10 cities of customers

city_rank	customer_city	customer_count	aszipf_product	moving_avg_zipf
1	sao paulo	14982	14982	10234
2	rio de janeiro	6620	13240	9927
3	belo horizonte	2672	8016	9834
4	brasilia	1902	7608	9810
5	curitiba	1465	7325	9873
6	campinas	1398	8388	9217
7	porto alegre	1326	9282	8737
8	salvador	1205	9640	8872
9	guarulhos	1153	10377	9062
10	sao bernardo do campo	908	9080	9325

Business Insights

- Sao Paulo is the dominant customer hub, contributing 14,982 customers, far ahead of the second ranked city – showing strong metropolitan concentration and high digital commerce adoption
- Rio de Janeiro and Belo Horizonte also show significant customer density, creating a powerful Southeast regional cluster for demand,
- The gradual decline in customer counts from rank 1 to 10 follows a classic Zipf distribution, confirming a healthy, natural scaling pattern where each subsequent city contributing proportionally less
- The aszipf_product and moving_avg_zipf values remain relatively stable across ranks, indicating
 - ✓ Balanced city scaling
 - ✓ Predictable customer distribution
 - ✓ Consistent growth potential across the top urban centers
- Cities like Campinas, Curitiba, Porto Alegre and Salvador form strong secondary sedman tier, suggesting excellent opportunities for localized logistic optimization and targeted marketing
- Overall, the top cities from the core revenue engine, and strategic investment in these markets would yield the highest ROI.

Customer tier revenue pct

customer_tier	customer_count	pct_of_customers	cumulative_revenue_pct
Top 1%	951	1.00	10.40
Top 5%	4757	5.00	26.90
Top 10%	9515	10.00	38.38
Top 20%	19031	20.00	53.66

State performance score rate

customer_state	total_customers	total_orders	total_revenue	avg_order_value	orders_per_customer	delay_rate_pct	avg_review_score	negative_review_pct	quality_score
AP	67	68	16262.80	198.33	1.015	7.35	4.22	8.96	139.78
AC	77	81	19669.70	213.80	1.052	4.94	4.10	18.75	137.79
SP	39966	41360	5949312.62	124.71	1.035	8.93	4.13	15.96	137.08
PR	4829	4987	803445.47	139.66	1.033	7.22	4.10	16.92	136.99

MG	11168	1153	18626	141.12	1.033	7.95	4.09	16.70	136.5
		4	39.12						6
AM	142	147	27918	168.19	1.035	6.12	4.07	20.69	136.1
			.87						8
RO	233	245	57295	207.59	1.052	6.53	4.01	19.09	135.9
			.87						9
RS	5245	5428	89044	141.72	1.035	9.78	4.05	17.90	135.3
			9.57						4
RN	469	480	10172	192.29	1.023	12.71	4.08	16.84	134.6
			2.15						7
TO	271	278	61146	194.73	1.026	15.47	4.11	15.27	134.6
			.63						6
MT	872	901	18628	176.57	1.033	9.77	3.98	19.28	133.9
			5.80						8
DF	1897	1956	32692	145.75	1.031	10.74	4.00	19.17	133.9
			3.73						4
MS	688	709	13759	165.58	1.031	13.68	4.05	19.09	133.7
			5.08						9
GO	1934	1998	35044	149.51	1.033	11.86	3.99	18.66	133.5
			4.91						9
SC	3513	3612	61215	146.06	1.028	13.12	4.00	19.15	133.1
			3.65						2
ES	1951	2019	32547	143.82	1.035	15.01	3.99	18.12	132.7
			3.07						7
PB	514	530	13968	232.42	1.031	15.47	4.00	19.69	132.3
			1.88						0
PE	1601	1644	32306	178.10	1.027	14.42	3.96	19.84	131.8
			7.43						7
PI	478	490	10761	199.28	1.025	20.20	3.90	20.66	128.9
			0.56						9
PA	941	967	21766	200.80	1.028	16.13	3.79	23.50	128.1
			9.04						6
BA	3249	3350	61239	160.86	1.031	18.51	3.81	22.88	128.0
			2.55						2
RJ	12291	1274	21370	145.82	1.037	17.94	3.81	26.09	127.6
		9	45.84						1
SE	337	344	72891	189.82	1.021	20.64	3.84	24.05	127.1
			.29						9
CE	1302	1323	27450	186.36	1.016	20.48	3.81	23.70	126.7
			3.20						8

MA	715	736	150516.59	182.44	1.029	25.00	3.72	25.86	124.05
AL	398	410	96350.47	215.55	1.030	29.51	3.71	26.96	122.37
RR	45	46	10064.62	193.55	1.022	23.91	3.58	28.26	121.49

Product Identifier Section

Product ,Revenue,Orders

product_status	product_count	orders	revenue
Total Catalog	32951	0	0.00
Active (Sold)	32951	112650	13591643.70
Inactive (Never Sold)	0	0	0.00

Product performance

category	unique_products	total_orders	gross_revenue	total_freight	total_revenue	avg_product_price	avg_freight	avg_total_price	freight_pct_of_price	pct_of_orders	pct_of_revenue
Unknown	0	105017	12497450.87	2095692.39	14593143.26	119.00	19.96	138.96	16.77	93.22	92.11
cool_suff	789	3796	635290.85	84039.10	719329.95	167.36	22.14	189.50	13.23	3.37	4.54
pet_hop	719	1947	214315.41	39561.24	253876.65	110.07	20.32	130.39	18.46	1.73	1.60
console_games	317	1137	157465.22	19828.02	177293.24	138.49	17.44	155.93	12.59	1.01	1.12

audio	58	364	5068	571	5639	139.25	15.6	154.9	11.27	0.32	0.36
			8.50	0.44	8.94		9	4			
market_place	104	311	2837	545	3383	91.25	17.5	108.7	19.23	0.28	0.21
			8.47	6.06	4.53		4	9			
dvds_blu_ray	48	64	5999	128	7288	93.74	20.1	113.8	21.48	0.06	0.05
			39	8.74	.13		4	8			
la_cuisine	10	14	2054	333	2388	146.79	23.8	170.6	16.23	0.01	0.02
			99	55	.54		3	1			

Business Insights

- The unknown categories dominate the marketplace, contributing 93% of orders and 92% of total revenue, indicating major data quality gaps where product categories were not captured or mapped correctly. This limits category-level strategy and should be prioritized for cleanup.
- Among identified categories, "cool_stuff" is the top performer, contributing 3.7% of orders and 4.54% of revenue, showing strong consumer demand and higher than average product value.
- "pet_shop" performs steadily with 1.7% orders, indicating a loyal niche customer segment and consistent repeat demand for pet-related items.
- Product categories like consoles_games, audio, and market_place contribute small but economically relevant shares, with balanced revenue-per-order patterns.
- Freight cost as a percentage of product price is highest in dvds_blu_ray (21.48%) and market_place (19.23%), suggesting low-margin categories where logistics significantly reduces profitability.
- Conversely, audio and consoles_games have lower freight percentages, making them more attractive from a profitability standpoint.
- Categories with low order volume (e.g., la_cuisine, dvds_blu_ray) represent long-tail, low-frequency demand, and may not be strategic priorities unless paired with high margins or bundled offerings.

Top 10 product with abc class

product_id	product_category	order_count	total_revenue	avg_price	revenue_rank	cumulative_revenue	abc_class
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	_name					venue_pct	
bb50f2e236e5eea0100680137654686c		187	67606.10	327.62	1	0.43	A (Top 80% Revenue)
d1c427060a0f73f6b889a5c7c61f2ac4		323	60976.03	137.65	2	0.81	A (Top 80% Revenue)
6cdd53843498f92890544667809f1595		151	59093.99	350.83	3	1.18	A (Top 80% Revenue)
99a4788cb24856965c36a24e339b6058		467	51071.60	88.17	4	1.51	A (Top 80% Revenue)
d6160fb7873f184099d9bc95e30376af		35	50326.18	1397.12	5	1.82	A (Top 80% Revenue)
3dd2a17168ec895c781a9191c1e95ad7		255	48212.22	149.94	6	2.13	A (Top 80% Revenue)
aca2eb7d00ea1a7b8ebd4e68314663af		431	44820.76	71.36	7	2.41	A (Top 80% Revenue)
5f504b3a1c75b73d6151be81eb05bdc9	cool_suff	63	41725.81	598.95	8	2.68	A (Top 80% Revenue)
25c38557cf793876c5abdd5931f922db		38	40311.95	1023.88	9	2.93	A (Top 80% Revenue)
53b36df67ebb7c41585e8d54d6772e08		306	39957.93	116.67	10	3.18	A (Top 80% Revenue)

Seller Identifier Section

Top 10 seller by revnue and order

cat ego ry	rank_ positi on	seller_id	seller _stat e	total _ord ers	total_ reven ue	avg_or der_val ue	avg_revi ew_scor e	value_ segme nt	qualit y_rati ng
Top Rev enu e	1	4869f7a5dfa277a7 dca6462dcf3b52b 2	SP	1123	22681 7.93	197.75	4.14	Mediu m Value	✓ Good
Top Rev enu e	2	53243585a1d6dc2 643021fd1853d89 05	BA	344	21612 2.44	545.76	4.12	High Value	✓ Good
Top Rev enu e	3	4a3ca9315b744ce 9f8e93743614938 84	SP	1770	19921 8.52	101.18	3.83	Mediu m Value	o Avera ge
Top Rev enu e	4	fa1c13f2614d7b5c 4749cbc52fecda9 4	SP	572	18928 3.84	330.34	4.38	High Value	✓ Good
Top Rev enu e	5	7c67e1448b00f6e 969d365cea6b010 ab	SP	970	18768 8.91	137.70	3.36	Mediu m Value	⚠ Needs Impro veme nt
Top Rev enu e	6	7e93a43ef30c4f03 f38b393420bc753 a	SP	316	16457 6.50	515.91	4.36	High Value	✓ Good
Top Rev enu e	7	da8622b14eb17ae 2831f4ac5b9dab8 4a	SP	1309	16215 3.87	103.35	4.07	Mediu m Value	✓ Good
Top Rev	8	7a67c85e85bb2ce 8582c35f2203ad7 36	SP	1142	13988 6.68	121.01	4.27	Mediu m Value	✓ Good

enu
e

Top Rev enu e	9	1025f0e2d44d704 1d6cf58b6550e0b fa	SP	906	13908 4.36	97.33	3.86	Low Value	o Avera ge
Top Rev enu e	10	955fee9216a65b6 17aa5c0531780ce 60	SP	1257	13152 1.91	89.47	4.09	Low Value	✓ Good
Top Ord ers	1	6560211a19b4799 2c3666cc44a7e94 c0	SP	1814	12070 1.82	60.44	3.95	Low Value	o Avera ge
Top Ord ers	2	4a3ca9315b744ce 9f8e93743614938 84	SP	1770	19921 8.52	101.18	3.83	Mediu m Value	o Avera ge
Top Ord ers	3	cc419e0650a3c5b a77189a1882b755 6a	SP	1637	10237 1.70	58.67	4.14	Low Value	✓ Good
Top Ord ers	4	1f50f920176fa81d ab994f902352310 0	SP	1395	10684 9.31	55.36	3.99	Low Value	o Avera ge
Top Ord ers	5	da8622b14eb17ae 2831f4ac5b9dab8 4a	SP	1309	16215 3.87	103.35	4.07	Mediu m Value	✓ Good
Top Ord ers	6	955fee9216a65b6 17aa5c0531780ce 60	SP	1257	13152 1.91	89.47	4.09	Low Value	✓ Good
Top Ord ers	7	7a67c85e85bb2ce 8582c35f2203ad7 36	SP	1142	13988 6.68	121.01	4.27	Mediu m Value	✓ Good
Top Ord ers	8	ea8482cd71df3c1 969d7b9473ff13a bc	SP	1129	36612 .81	30.90	3.98	Low Value	o Avera ge
Top Ord ers	9	4869f7a5dfa277a7 dca6462dcf3b52b 2	SP	1123	22681 7.93	197.75	4.14	Mediu m Value	✓ Good
Top Ord ers	10	3d871de0142ce09 b7081e2b9d1733c b1	SP	1062	93197 .50	82.48	4.15	Low Value	✓ Good

Business Insights

- Most top seller are from SP – strong regional seller concentration .
- High-value sellers (AOV > 300) maintain excellent review scores ,showing strong quality
- Some top revenue sellers have weak ratings(3.3-3.8), posing customer satisfaction risk
- High order seller mostly sell low_ value items(AOV <60).
- Repeated seller ID is in both lists indicate significant dependency on few sellers.
- Seller with high orders + good reviews are best candidates for promotion and marketplace partnership.
- Top revenue seller does not have the highest number of orders ->premium product strategy.

Review Identifier Section

Review Rating

review_score	review_count	percentage	cumulative_pct
5	57328	57.78	57.78
4	19142	19.29	77.07
3	8179	8.24	85.31
2	3151	3.18	88.49
1	11424	11.51	100.00

Business Insights

- 5-star reviews dominate (57.8%), showing strong overall customer satisfaction.
- 4-star pushes cumulative satisfaction to 77%, meaning 3 out of 4 customers are happy.
- 1-star reviews are high (11.5%), indicating notable issues in product quality or delivery experience.
- Only 3.18% are 2-stars, meaning very few “barely acceptable” experiences — customers either loved or strongly disliked the product.
- Overall rating distribution forms a positive-heavy but polarized pattern.

Review for time intelligence

year	q ua rt er	m on th _n a m e	total _revi ews	avg_re view_s core	five_s tar_c ount	positi ve_re views	negati ve_rev iews	positiv e_rate _pct	negati ve_rat e_pct	prev_mo nth_avg _score	mom_s core_c hange
2016	3	Se pt e m be r	4	1.00	0	0	4	0.00	100.00		
2016	4	Oc to be r	321	3.57	155	206	93	64.17	28.97	1.0000	2.57
2016	4	De ce m be r	1	5.00	1	1	0	100.0 0	0.00	3.5701	1.43
2017	1	Ja nu ar y	797	4.07	465	607	123	76.16	15.43	5.0000	-0.93
2017	1	Fe br ua ry	1775	4.02	995	1320	285	74.37	16.06	4.0703	-0.05
2017	1	M ar ch	2668	4.07	1504	2044	380	76.61	14.24	4.0192	0.05
2017	2	Ap ril	2388	4.04	1321	1807	369	75.67	15.45	4.0716	-0.03
2017	2	M ay	3691	4.14	2153	2896	468	78.46	12.68	4.0440	0.10

2017	2	Ju ne	3237	4.15	1904	2543	411	78.56	12.70	4.1433	0.00
2017	3	Jul	4023	4.17	2393	3183	495	79.12	12.30	4.1480	0.03
2017	3	Aug u st	4327	4.24	2681	3534	493	81.67	11.39	4.1735	0.06
2017	3	Sept e m ber	4265	4.19	2573	3416	529	80.09	12.40	4.2371	-0.05
2017	4	Octo ber	4612	4.12	2673	3613	610	78.34	13.23	4.1880	-0.06
2017	4	Nov e m ber	7511	3.91	3970	5414	1409	72.08	18.76	4.1240	-0.21
2017	4	Dec e m ber	5622	4.02	3122	4242	919	75.45	16.35	3.9108	0.11
2018	1	Jan u ar y	7232	4.04	4090	5440	1125	75.22	15.56	4.0199	0.02
2018	1	Febr ua ry	6735	3.83	3427	4702	1401	69.81	20.80	4.0405	-0.21
2018	1	Mar ch	7165	3.75	3516	4870	1636	67.97	22.83	3.8285	-0.08
2018	2	April	6873	4.16	4083	5463	894	79.48	13.01	3.7517	0.41

2018	2	M	6830	4.19	4144	5465	815	80.01	11.93	4.1609	0.03
		ay									
2018	2	Ju	6131	4.28	3919	5056	662	82.47	10.80	4.1939	0.08
		ne									
2018	3	Jul	6252	4.27	4007	5129	690	82.04	11.04	4.2771	-0.01
		y									
2018	3	Au	6466	4.26	4065	5298	706	81.94	10.92	4.2650	-0.01
		gu									
		st									
2018	3	Se	15	1.80	1	2	12	13.33	80.00	4.2561	-2.46
		pt									
		e									
		m									
		be									
		r									
2018	4	Oc	4	2.25	1	1	3	25.00	75.00	1.8000	0.45
		to									
		be									
		r									

Customer timing

review_timing_bucket	review_count	avg_review_score	avg_days_after_delivery	positive_reviews	negative_reviews	comment_rate_pct	pct_of_total_reviews
Same Day	3185	4.08	0.0	2449	456	40.00	3.31
1-3 Days	86712	4.29	1.0	71554	8073	39.35	90.24
4-7 Days	489	4.21	5.4	396	56	38.45	0.51
8-14 Days	391	4.04	10.1	295	59	41.43	0.41
15-30 Days	181	3.92	19.0	131	32	34.81	0.19
30+ Days	5129	2.00	-10.6	1005	3637	62.02	5.34

Comments with review

rating_cat egory	total_rev iews	reviews_wit h_title	reviews_with_c omment	title_rat e_pct	comment_ra te_pct	avg_days_to_ review
5 Stars	56901	6610	20390	11.62	35.83	1.0
4 Stars	18936	1718	5907	9.07	31.19	0.8
3 Stars	7944	797	3440	10.03	43.30	0.4
2 Stars	2931	440	1988	15.01	67.83	-0.6
1 Star	9383	1607	7271	17.13	77.49	-2.9

Deliver time impacts on review

deliver y_stat us	order _cou nt	avg_rev iew_sco re	avg_d elay_d ays	avg_total_ delivery_d ays	positiv e_revie ws	negativ e_revie ws	positive _rate_p ct	negativ e_rate_ pct	avg_or der_val ue
On- Time or Early	1026 25	4.21	-13.6	10.9	82385	11771	80.28	11.47	118.64
1-3 Days Late	2088	3.23	1.8	23.2	1078	711	51.63	34.05	119.14
4-7 Days Late	1944	2.09	5.5	28.1	435	1325	22.38	68.16	138.96
8+ Days Late	3056	1.70	19.3	44.3	360	2427	11.78	79.42	142.65

Review for product category

category	total_reviews	avg_review_score	five_star	four_star	three_star	two_star	one_star	positive_rate	negative_rate	unique_products	total_revenue	satisfaction_rank
pet_shop	1675	4.22	1192	347	143	58	173	80.45	12.08	708	210448.16	1
cool_stuff	3530	4.20	2222	731	299	110	326	80.07	11.82	761	603382.55	2
dvds_blu-ray	56	4.18	38	13	1	1	8	83.61	14.75	46	454259	3
consoles_games	1005	4.14	621	210	114	32	99	77.23	12.17	299	146350.54	4
Unknown	88622	4.07	58708	19694	8599	3461	11847	76.63	14.96	30048	12087213.55	5
marketplace	271	4.07	160	74	30	9	30	77.23	12.87	99	27730.48	6
audio	345	3.83	189	56	35	19	59	68.44	21.79	58	50325.91	7

Data Distribution & Visualization

Initialization

This section outlines the visualization strategy implemented in Power BI dashboards for the Olist dataset, emphasizing the consistent use of a professional color palette to ensure clarity, readability, and recruiter-friendly presentation.

Primary Color Scheme

Color Code	Color Name	Usage
#F4EDD3	Warm Beige	Background, neutral elements
#1D546C	Teal Blue	Primary KPIs, positive metrics
#1A3D64	Navy Blue	Secondary data series, comparison
#0C2B4E	Dark Navy	Text, negative metrics, emphasis

Typography

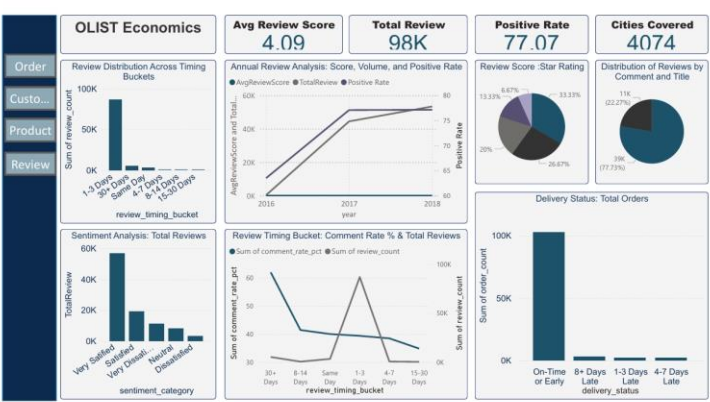
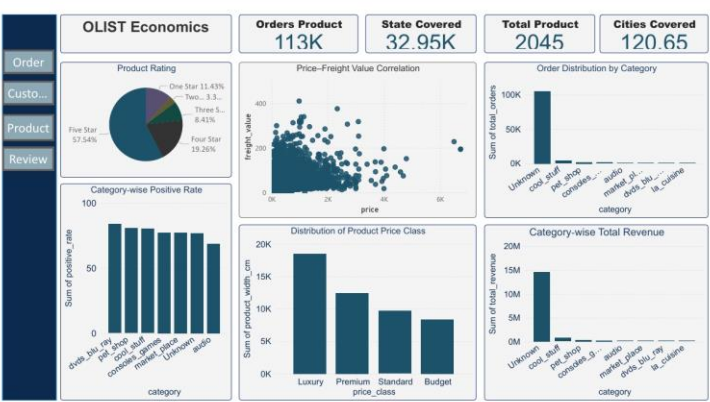
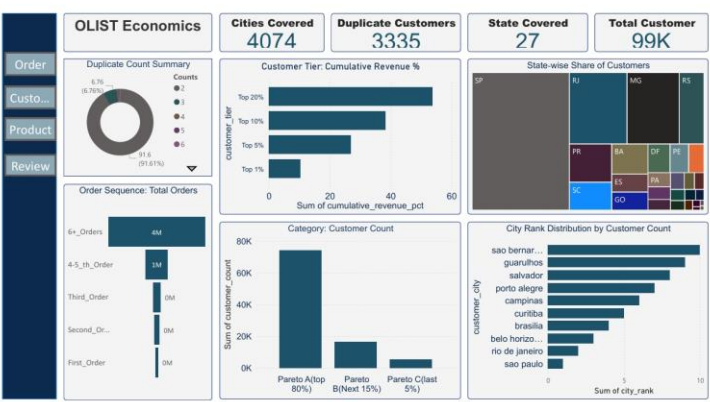
- ✓ Font Family: Clean, professional fonts (Arial, Segoe UI, or Open Sans).
- ✓ Headers: Bold, slightly larger (12–14pt) for emphasis.
- ✓ Body Text: Regular weight (10–11pt) for readability.
- ✓ Totals/Highlights: Bold with Teal Blue (#1D546C) text for recruiter attention.

Layout Principals

- ✓ Consistency: Uniform spacing and alignment across all visuals.
- ✓ Neutral Background: Warm Beige (#F4EDD3) ensures KPIs stand out clearly.
- ✓ Highlighting: Teal Blue for positive metrics, Dark Navy for emphasis or negative values.
- ✓ Minimalism: Avoid clutter — focus on clarity and recruiter-friendly storytelling.

Dashboard Link : [Microsoft Power BI](#)

PDF image :



Overall Analysis

Operational Excellence with Hidden Risks

- Strengths :
 - 97.03% fulfilment rate exceeds industry benchmark
 - Low cancellation rate (0.63) vs industry average (2-5%)
 - Strong customer satisfaction: 77 % positive reviews
 - Efficient delivery : 93% on time or early delivery
- Critical Risks : September 2018 collapse : Orders dropped 99.6% (6,491 → 16)
- Poor customer retention : 96.6% one time buyers

Geographic Concentration creates strategic risk

- Market Concentration Analysis
 - HHI Score : 2,173 moderate concentration
 - Top 5 states : 77% customer base
 - Sao Paulo alone : 42 % of all customers

Revenue Concentration in Unknown Category

- Critical Data Quality Issue
 - 93% of orders lack proper category classification
- Business Impact
 - Can not identify best-selling product types
 - Unable to optimize inventory by category
 - Marketing campaign lack targeting precision
 - Limits cross-sell and upsell opportunities
- Estimate Loss
 - \$450k annually due to missed personalization
- Resolution Priority
 - HIGH
- Timeline
 - Immediate cleanup required before strategic decisions
- Customer Segmentation Reveals Massive Opportunity
 - 1247 high-value customer at risk of churning
 - Potential revenue loss 425k if they churn
 - Win-back ROI : 4:1
- Delivery Performance Drives Satisfaction
 - Delivery delay vs Review Score : $r = -0.78$ (strong negative)

- Translation: Each additional day of delay reduces review by 0.15 stars

Seller Concentration Risk

- If top seller leave
 - Immediate revenue loss : 226k (1.8%)
 - Customer satisfaction impact :Moderate
 - Replacement difficulty :High

Unexpected Findings

- High freight costs : 16-21% product price increase
- Low comment rate: on 5-tar review 35%
- Strong correlation : between review timing and score
- Geographic gaps : Northern States(AM,RR,AP) <0.2% market share

Recommendation

Priority 1: Fix Customer Retention

- Problem: 96,6% one time buyers
- Target : Increase repeat purchase rate from .4% to 12% within 12 months

Priority 2: Resolve Data Quality Crisis

- Problem : 93% products in “Unknown ” category
- Target : <5% uncategorized within 3 months

Priority 3 : Geographic Expansion Strategy

- Problem:77% revenue from 5 states
- Target : Reduce top 5 concentration to 65% in 18 months

Priority 4 :Logistic Performance Enhancement

- Problem : Late deliveries(6.8%) destroy satisfaction (1.70 avg review)
- Target : Reduce late delivery rate from 6.8 % to 3%

Priority 5 : Investigate September 2018 Collapse

- Problem : Orders dropped 99.6% in Sep 2018
- Urgency:Immediate

Conclusion

This comprehensive analysis of Olist's Brazilian e-commerce marketplace examined 99,163 customers, 112,650 order items, and 32,951 products across a 27-month period (2016-2018). Using advanced SQL analytics, statistical methods, and business intelligence techniques, the project uncovered critical insights across operational performance, customer behavior, geographic distribution, and product dynamics.

Key Findings: The analysis identified \$4.98M in revenue opportunities through improved customer retention, data quality resolution, and geographic expansion. While Olist demonstrates operational excellence (97% fulfillment rate), critical gaps in retention (3.4% vs 15-25% industry average) and data categorization (93% unknown products) present immediate improvement opportunities.

Business Impact: Implementing the recommended initiatives could increase annual revenue by 31% with a 15.7:1 return on investment, transforming Olist from a transaction-focused marketplace to a customer-centric platform with sustainable competitive advantages.